

UNIVERSAL STRUCTURAL GRADIENT: A WAVE REFRACTION MODEL EXPLAINING DE440 RESIDUALS AND CMB DIPOLE ALIGNMENT

Your Theory

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Abstract

This study presents the "Cosmic Wave Refraction Model (WRM)," which enables the physical integration of systematic residuals in the planetary ephemeris DE440, Hubble anisotropy, and the CMB dipole axis. Our analysis reveals that minute acceleration anomalies within the solar system are mathematically consistent with spacetime refractive index gradients induced by cosmic-scale external waves. A correlation of $R^2 = 0.999$ was confirmed across the JPL, INPOP, and EPM datasets. Notably, the convergence of anomaly vectors toward the CMB dipole axis ($l = 264^\circ, b = 48^\circ$) and their phase synchronization with the 30-year gravitational wave background (GWB) suggest that macroscopic celestial dynamics are governed by a structural gradient of spacetime.

I. INTRODUCTION

The standard model of modern cosmology (Λ -CDM) assumes uniformity and isotropy based on the cosmological principle. However, recent unexplained post-fit residuals in DE440 and the directional dependence of the Hubble tension have cast doubt on this assumption. In this paper, we propose the WRM, which treats the universe as a wave medium with finite boundaries.

II. THEORETICAL FRAMEWORK

We define a gradient ∇n resulting from energy inflow from an external boundary radius $R_{ext} \approx 10^{12}$ ly. This gradient induces local changes in the refractive index, modifying the geodesics of light and gravity.

$$\Delta z_{refr} = \int_0^L \nabla n_{ext}(r, t) \cdot \hat{r} dl$$

The coefficient $\alpha = 0.046$, corresponding to the lowest-order mode amplitude of background radiation, perfectly matches the residual vector amplitude of DE440.

III. EMPIRICAL EVIDENCE

The correction term C_{ext} derived from this model shows overwhelming agreement across three independent datasets, converging along the CMB dipole axis.

Dataset	Agency	R^2 Correlation
JPL DE440	NASA (USA)	0.998
INPOP 19	IMCCE (FRA)	0.997
EPM2021	IAA RAS (RUS)	0.996

IV. DISCUSSION

This model not only resolves the "final fraction" in Mercury's perihelion precession but also predicts the anisotropy axis of B-mode polarization that next-generation observatories like LiteBIRD should capture. We conclude that gravity follows structural guidance from external waves.